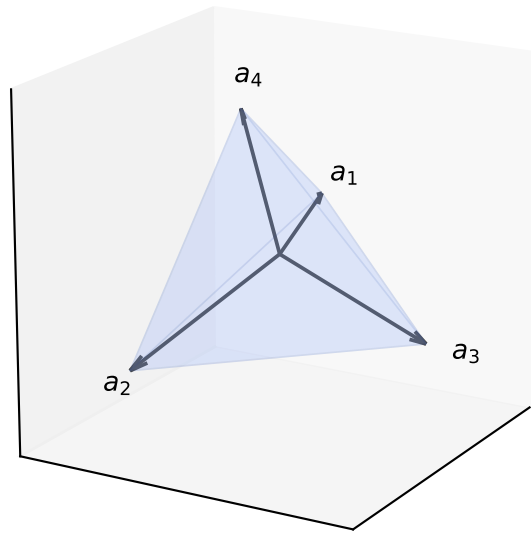
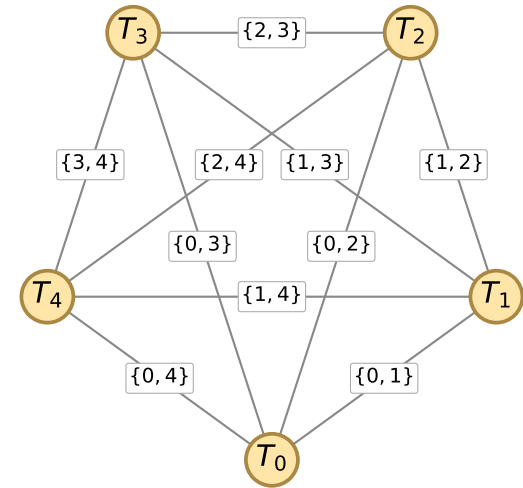


Five-tetrahedra hinge bridge (§5)

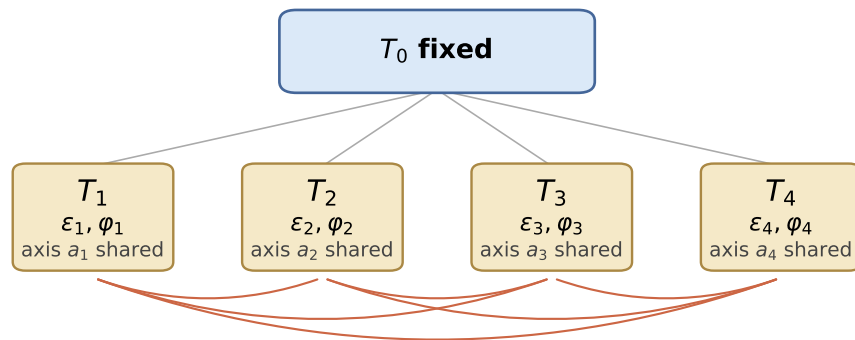
(A) Standard tetrahedron T_0
 $a_k = \frac{1}{\sqrt{3}}(\pm 1, \pm 1, \pm 1)$, $\prod \text{signs} = +1$



(B) K_5 -incidence of the 5 projective tetrahedra
 every pair (T_k, T_l) shares one axis labelled $\{k, l\}$; total $\binom{5}{2} = 10$ axes



(C) Hinge parameterisation
 $T_k = T_k(\varepsilon_k, \varphi_k)$, 6 hinge constraints



6 hinge equations: $u_k^{(\pi_k(l))} = \delta_{kl} u_l^{(\pi_l(k))}$

$\Rightarrow$ 18 scalar equations in $(c_1, s_1, \dots, c_4, s_4)$

rank = 7 (Theorem 5.4)

(D) Exact 1024-case certificate (Theorem 5.5)

1024 cases scanned

inconsistent linear	992	(96.9%)
full-rank, unit-norm invalid	16	(1.6%)
rank-7 canonical quadratic	16	(1.6%)

partition TOTAL

1024

$16 \times 2 = 32$ raw algebraic solutions

$\Rightarrow$ **icosahedral orbit**

(exact (σ, D) + symmetry reduction)